



The Complete DHCP Picture - Everything Combined

Let me stitch together everything we discussed into one cohesive understanding of DHCP.

What is DHCP and Why It Exists

DHCP = Dynamic Host Configuration Protocol

The fundamental problem:

- You have dozens/hundreds of devices
- Each needs a unique IP address
- Manual configuration = nightmare
- Risk of IP conflicts

DHCP's solution:

- Automatic IP assignment
- Centralized management
- No conflicts
- Lease-based (IPs can be reused)

What DHCP gives you:

1. IP address (e.g., 192.168.1.50)
2. Subnet mask (e.g., 255.255.255.0)
3. Default gateway (e.g., 192.168.1.1)
4. DNS servers (e.g., 8.8.8.8)
5. Lease time (e.g., 24 hours)

The 4-Way Handshake (DORA Process)

Step 1: DISCOVER (Broadcast)

Your Device:

```
Just connected to network
IP: 0.0.0.0 (no IP yet!)

Sends: DHCP DISCOVER
To: 255.255.255.255 (broadcast)
MAC: FF:FF:FF:FF:FF:FF (broadcast)

Message: "I need an IP! Anyone?"
```

Why broadcast?

- Don't know which device is DHCP server
- Don't have IP address yet (0.0.0.0)
- All devices receive it, but only DHCP servers respond

Step 2: OFFER (Unicast)

DHCP Server(s):

```
Router 1:
"I can offer: 192.168.1.100"
Sends: DHCP OFFER
To: Your device's MAC (unicast)

Router 2 (if exists):
"I can offer: 192.168.1.200"
Also sends OFFER
```

Why unicast?

- Server knows your MAC address (from DISCOVER)
- Responds specifically to you
- Other devices don't need to see this

Multiple servers:

- Networks may have 2+ DHCP servers (redundancy)
- You might receive multiple OFFERs
- You pick one (usually first to arrive)

Step 3: REQUEST (Broadcast)

Your Device:

```
Received offer(s), choosing one
Sends: DHCP REQUEST
To: 255.255.255.255 (broadcast!)
Content: "I accept Router 1's
         offer of 192.168.1.100
         Server ID: 192.168.1.1"
```

Why broadcast (not unicast)?

This is critical! Even though you know which server you're accepting from, you broadcast because:

1. **Tell rejected servers** - Other DHCP servers need to know you rejected their offers so they can release those IPs back to their pools
2. **Prevent IP waste** - Without broadcast, rejected servers keep IPs reserved
3. **Network change detection** - Helps detect if you moved to different network

Router 1: "They chose me! I'll send ACK" ✓

Router 2: "They chose Router 1, I'll release 192.168.1.200" ✓

Why not multicast?

- You don't have IP address yet (0.0.0.0)
- Requires network configuration you don't have
- Switches would need IGMP snooping
- Broadcast is simpler and universally supported
- "Waste" is negligible (DHCP is rare, packets small)

Step 4: ACK (Unicast)

DHCP Server (Router 1):

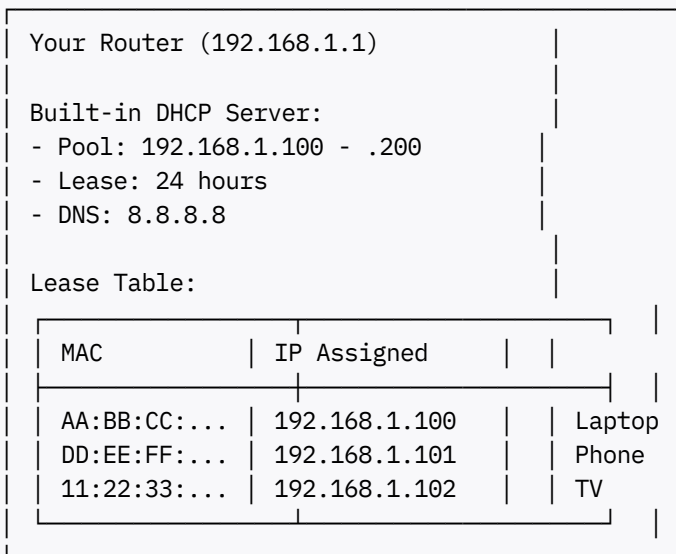
```
"Confirmed! Here's everything:"
IP: 192.168.1.100
Subnet: 255.255.255.0
Gateway: 192.168.1.1
DNS: 8.8.8.8, 8.8.4.4
Lease: 86400 seconds (24 hours)
"हां भाई ले ले, जी ले अपनी ज़िंदगी!" |
```

Your device now:

- Configures IP: 192.168.1.100
- Sets gateway: 192.168.1.1
- Uses DNS: 8.8.8.8
- Can access internet! ☑

DHCP in Different Contexts

Home Network

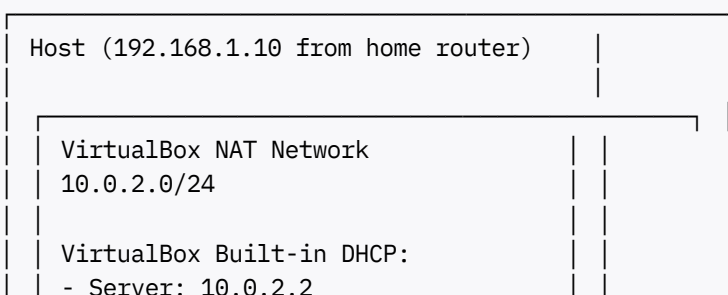


Single DHCP server (router)

- No multiple OFFERs
- Simple setup
- REQUEST still broadcast (protocol requirement)

VirtualBox Networks

NAT Mode - VirtualBox DHCP Server



```
- Pool: 10.0.2.15 - 10.0.2.254
- Gateway: 10.0.2.2
```

```
VM
Gets: 10.0.2.15
DHCP from VirtualBox, not router
```

VM's DHCP process:

1. VM sends DISCOVER
2. VirtualBox intercepts it (never reaches home router)
3. VirtualBox DHCP responds with OFFER (10.0.2.15)
4. VM sends REQUEST (broadcast in VM's network)
5. VirtualBox sends ACK

Home router never sees this!

Bridged Mode - Uses Home Router's DHCP

```
Host (192.168.1.10)
```

```
VM in Bridged Mode
Sends DHCP DISCOVER
```

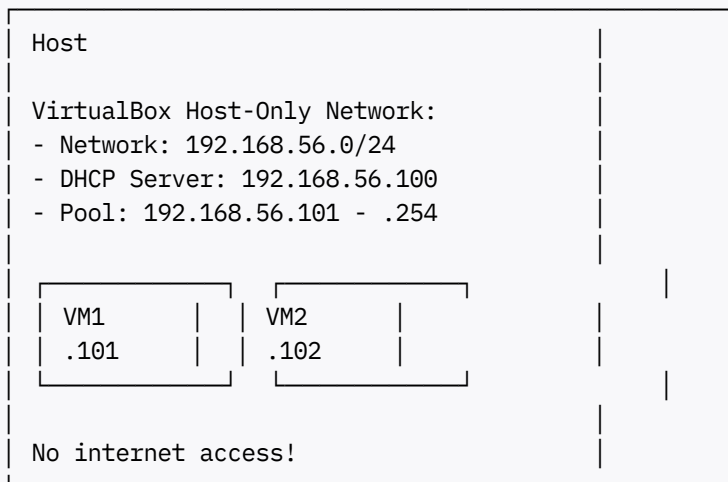
(Bridge forwards to network)
↓

```
Home Router
Sees VM as separate device!
Gives: 192.168.1.101
```

VM gets real IP from home router

- Acts like physical device
- Uses same DHCP as host
- Router sees two devices (host + VM)

Host-Only Mode - Separate VirtualBox DHCP



Isolated DHCP

- VMs get IPs from VirtualBox
- Can talk to each other and host
- No internet (no gateway configured)

DHCP Lease Management

Initial Lease (24 hours example)

```
Day 1, 10:00 AM - Device connects
↓
DORA handshake
↓
IP: 192.168.1.100
Lease expires: Day 2, 10:00 AM
```

Lease Renewal (Automatic)

```
Day 2, 5:00 AM (50% of lease time)
Device: "DHCP REQUEST (unicast to server)"
      "Can I keep 192.168.1.100?"
Server: "ACK - Yes! Another 24 hours"
New expiry: Day 3, 5:00 AM

Day 2, 8:45 AM (87.5% of lease time, if above failed)
Device: "DHCP REQUEST (broadcast)"
      "Can anyone renew my lease?"
Server: "ACK - Sure!"
```

Key differences in renewal:

- REQUEST is **unicast** (knows server address)
- No DISCOVER needed
- Usually gets same IP
- If renewal fails, starts fresh DORA

Lease Expiry

If device is off when lease expires:

Day 3, 10:00 AM - Device turns on
Device IP: Invalid (lease expired)

↓

Starts fresh DORA process

↓

Might get same IP (if available)
Might get different IP

Static vs Dynamic IP

Dynamic IP (DHCP) - What We Discussed

Pros:

- ✓ Automatic configuration
- ✓ No IP conflicts
- ✓ Central management
- ✓ IPs reused efficiently
- ✓ Easy for guests/temp devices

Cons:

- ✗ IP can change
- ✗ Can't rely on IP for servers
- ✗ Depends on DHCP server availability

Use for:

- Laptops, phones, tablets
- Guest devices
- Most client devices

Static IP (Manual Configuration)

Manual settings:

IP: 192.168.1.50

Subnet: 255.255.255.0

Gateway: 192.168.1.1

DNS: 8.8.8.8

Pros:

- ✓ IP never changes
- ✓ Reliable for servers
- ✓ Port forwarding works
- ✓ No DHCP dependency

Cons:

- ✗ Manual configuration
- ✗ Risk of conflicts
- ✗ Hard to manage at scale

Use for:

- Servers (web, database)
- Printers
- Network equipment
- IoT hubs

DHCP Reservation (Best of Both)

Router configuration:

MAC AA:BB:CC:DD:EE:FF → Always gets 192.168.1.50

Device still uses DHCP!

But always gets same IP

Combines benefits:

- ✓ Automatic (DHCP)
- ✓ Consistent IP
- ✓ Central management

What Each Layer Sees

During DHCP Process

VM's perspective (NAT mode):

```
"I did DHCP handshake"  
"Got IP: 10.0.2.15"  
"Gateway: 10.0.2.2"
```

Host OS perspective (NAT mode):

```
"VirtualBox is running"  
"VirtualBox accessing network"  
"See VirtualBox's connections to internet"
```

Host OS perspective (Bridged mode):


```
"VirtualBox is running"  
"VirtualBox accessing NIC hardware"  
"Don't see VM's traffic (bypasses my stack)"
```

Router perspective (NAT mode):

```
"Host (192.168.1.10) is online"  
"Don't see any VM"
```

Router perspective (Bridged mode):

```
"Host (192.168.1.10) is online"  
"VM (192.168.1.101) is online" ← Separate device!
```

Port Usage in DHCP

DHCP Uses Specific Ports

```
DHCP Client → Server:  
Source Port: 68 (DHCP client)  
Dest Port: 67 (DHCP server)
```

```
DHCP Server → Client:  
Source Port: 67 (DHCP server)  
Dest Port: 68 (DHCP client)
```

How devices know to ignore DHCP broadcasts:

```
Your Laptop (not a DHCP server):  
Receives DHCP packet  
↓  
Checks: Destination port = 67  
↓  
"I'm not listening on port 67"  
↓  
Drop packet (in network driver, instant)  
↓  
CPU barely notices! (microseconds)
```

This is why broadcast "waste" doesn't matter!

Network Address Concepts

We learned network addresses earlier - here's how DHCP relates:

Network: 192.168.1.0/24

Reserved:

- 192.168.1.0 (network address) ✗ Not assigned by DHCP
- 192.168.1.255 (broadcast) ✗ Not assigned by DHCP

Router usually:

- 192.168.1.1 (gateway) ✗ Not in DHCP pool

DHCP Pool:

- 192.168.1.100 to 192.168.1.200 ✓ Available for assignment

DHCP respects reserved addresses!

Summary: The Complete DHCP Flow

1. Device connects to network
 - ├ Has no IP (0.0.0.0)
 - └ Needs configuration
2. DISCOVER (Broadcast)
 - ├ "Anyone got an IP for me?"
 - └ All devices hear, only DHCP servers respond
3. OFFER (Unicast)
 - ├ DHCP server(s) offer IP
 - └ Device receives one or more offers
4. REQUEST (Broadcast)
 - ├ "I'm taking this offer!"
 - ├ Broadcast so rejected servers know
 - └ They can release their offered IPs
5. ACK (Unicast)
 - ├ Server confirms
 - └ Gives IP + subnet + gateway + DNS + lease time
6. Device configured!
 - ├ Uses IP for lease duration
 - └ Renews automatically at 50% and 87.5%
7. Lease expires if device off
 - └ Restart DORA when device reconnects

Key Takeaways

- ✓ **DHCP automates IP assignment** - saves manual configuration
- ✓ **4-step DORA** - Discover, Offer, Request, Acknowledge
- ✓ **REQUEST is broadcast** - tells rejected servers to release IPs
- ✓ **Broadcast is better than multicast** - simpler, works without IP configuration
- ✓ **Different DHCP servers in VirtualBox modes:**
 - NAT: VirtualBox DHCP (10.0.2.x)
 - Bridged: Home router DHCP (192.168.1.x)
 - Host-Only: VirtualBox DHCP (192.168.56.x)
- ✓ **Leases are temporary** - renewed automatically, IPs reused when expired
- ✓ **Static vs Dynamic** - choose based on need (servers vs clients)
- ✓ **Ports 67/68** - how devices know to ignore DHCP broadcasts
- ✓ **Works with NAT/ports** - connects to our earlier VM networking discussion

The magic: DHCP makes networking "just work" without users needing to understand any of this! ☐